

Chemistry Ch 5.5: Bonding, Structure and Properties

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Ionic aur Covalent compounds ka muqabla

Khasoosiyat	Ionic	Covalent
Melting / boiling point	Ooncha	Neecha
Halat (kamre ke temp par)	Aam tor par thos	Aksar maaye ya gas
Bijli chalana	Pighli ya ghuli halat mein haan	Aam tor par nahin
Paani mein ghulna	Aam tor par ghul jate hain	Aksar nahin

Ionic compound thos halat mein bijli kyun nahin chalata

Bijli chalane ke liye charge ka **chalna** zaroori hai.

Thos halat mein ions apni jagah par **jakre** hue hain, hil nahin sakte. To bijli nahin chalti.

Pighla dein ya paani mein ghol dein to ions **azad** ho jate hain aur chal sakte hain. Ab bijli chalti hai.

Yani ions wohi hain, bas **harkat** ka farq hai. Ye sawal har saal aata hai.

5.5.1 Graphite aur 5.5.2 Diamond

Dono **khalis carbon** hain, magar banawat alag hai (ye aap Chapter 2 mein parh chuki hain).

Graphite: tehon mein, har carbon **teen** se jura, chautha electron **azad**. Is liye **naram** aur **bijli chalata** hai.

Uses: lubricant, electrode, pencil.

Diamond: har carbon **chaar** se jura, koi azad electron nahin. Is liye **sab se sakht** aur bijli **nahin** chalata.

Uses: kaatne ke auzaar, zewar.

Tezab bhi bijli chalate hain

Kitab khaas taur par likhti hai ke tezab paani mein **ions** dete hain, aur isi liye un ka mahlool bijli chalata hai. Wajah wohi hai: azad ions.

Quiz

1. Compare ionic and covalent compounds for melting point, state at room temperature, conduction of electricity and solubility in water.
2. Explain why an ionic compound does **not** conduct electricity when solid but **does** when molten or dissolved. Then give two uses each for graphite and diamond, with the reason behind each use.